Brian Jongwon Choi

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Nationality: USA

Interests Partial differential equations, Harmonic analysis, Numerical analysis, Dynamical systems, Uncertainty quantification

Education

Boston University
- PhD, Mathematics, MAY20
- Thesis (under *Mark Kon*): *Weighted Fourier Analysis and Dispersive Equations*

Boston College
- B.S., Magna Cum Laude, Mathematics (Major), Physics (Minor), MAY14
- Recipient of the Sally Award 2014 (awarded to the top graduating math major)
- Member of Pi Mu Epsilon (mathematics) and Sigma Pi Sigma (physics)
- Thesis (under *Dubi Kelmer*):
  An Algebraic Approach to Quantum Systems Using Finite Group Representation

Appointment

University of Tennessee at Chattanooga
- (Tenure-track) Assistant professor (Department of Mathematics), JUL25–present

United States Military Academy West Point
- Assistant Professor (Department of Mathematical Sciences), JUN23–JUN25
- Deputy Director for Center for Data Science, JUN23–JUN25

Southern Methodist University
- RTG Postdoctoral Fellow (Mentor: Alejandro Aceves), JAN21–JUN23

Boston University
- Research Assistant, JUN20–DEC20

Boston College
- Instructor, MATH1004 Finite Probability and Applications, SEP20–DEC20

Awards/Grants

Civilian Service Commendation Medal, Department of the Army SEP25
Faculty Research Funds (competitive internal grant), USMA, West Point 2024–2025
Science Policy Fellowship, Society for Industrial and Applied Mathematics 2025–2026
Korean Honor Scholarship, Consulate General of Korea in USA SEP19
DAAD RISE Scholarship, Paul-Erlich-Institut, Germany MAY14–AUG14
Sally Award, Department of Mathematics, Boston College MAY14
Gilman International Scholarship, Institute of International Education MAY12

Publications/ Preprints

Long-time continuum limits for twisted Schrödinger lattices (2026) arXiv:2609.34081 (submitted to *Journal of Differential Equations*)

(with Castrillon, Kon, Norton, Xu) Analytic regularity of strong solutions for the complexified stochastic nonlinear Poisson-Boltzmann equation *Computers & Mathematics with Applications* (2026), **204**, p.97-122, doi.org/10.1016/j.camwa.2025.12.003.

Nonexistence of traveling wave solutions in the fractional Rosenau-Hyman equation via homotopy perturbation method (2025) arXiv:2502.07810 (submitted to *Electronic Journal of Differential Equations*)

Nonlinear excitation of ground states on nonlocal lattices, *Discrete Contin. Dyn. Syst. Ser. B* (2025), doi.org/10.3934/dcdsb.2025162

(With Parcell, Starling) Optimizing performance of real-time detection and classification for military personnel and weapons *The ITEA Journal* (2024), **45**(3), DOI: 10.61278/itea.45.3.1005

Dynamics of periodic fractional discrete NLS in the continuum limit, *Applicable Analysis* (2025), **104**(14), doi.org/10.1080/00036811.2025.2557997.

(With Norton, Xu, Kon, Castrillon) Complex analyticity of the nonlinear Poisson-Boltzmann equation for the interface problem with random domains, *Numerische Mathematik* (2025), doi.org/10.1007/s00211-025-01509-z.

(With Marstaller, Aceves) Localization and global dynamics in the long-range discrete nonlinear Schrödinger equation (2023) arXiv:2309.11395 (under final revision: *Journal of Difference Equations and Application*)

(with Walton) Speed of propagation of fractional dispersive waves, *Communications on Pure and Applied Analysis* (2025), doi:10.3934/cpaa.2025094.

(with Aceves) Continuum limit of 2D fractional nonlinear Schrödinger equation *Journal of Evolution Equations* (2023), **23**(30), p.1-35, doi.org/10.1007/s00028-023-00881-3.

(with Aceves) Well-posedness of mixed nonlinear Schrödinger equation, *Partial Differential Equations in Applied Mathematics* (2022), **6**(c), p.1-11, 100406, doi.org/10.1016/j.padiff.2022.100406.

Multilinear weighted estimates and quantum Zakharov system, *Mathematical Modelling and Analysis* (2022), **27**(2), 342-359, doi.org/10.3846/mma.2022.15555.

Remark on the adiabatic limit of quantum Zakharov system, *Bulletin of the Malaysian Mathematical Sciences Society* (2022), **45**, 17111735, doi.org/10.1007/s40840-022-01272-6.

Small time behavior and summability for Schrödinger equation, *Graduate Journal of Mathematics* (2021), **6**(2), 9-21.

Current Projects

- From twisted DNLS to magnetic NLS: continuum limits and nodal standing waves
- Long-time uncertainty quantification with applications in 1. nonlinear phase transitions; 2. nonlinear dispersive phenomena.

University Teaching

Academic Semester Teaching, *University of Tennessee at Chattanooga*

- MATH 2560: Calculus III — SPRING27
- MATH 1730: Precalculus — FALL26
- MATH 3810: Quantum Information Science and Technology — FALL26

Academic Semester Teaching, *University of Tennessee at Chattanooga*

- MATH 1730: Precalculus; MATH 3000: Proof and logic — AY25–26

Academic Semester Teaching, *United States Military Academy*

- MA364 ENG. Math — SPRING25
- MA487 Real analysis II, MA365 ENG. Math, MA389 Indep. study — FALL24
- MA104 Calculus I-II (Combined) — SPRING24

- MA205 Multivariable calculus FALL23

Academic Semester Teaching, *Southern Methodist University* JAN21–MAY23
- MATH1337,1338,3302 Calculus I-III, respectively
- STAT2331 Introduction to Statistical Methods

Academic Semester Teaching, *Boston College*
- MATH1004 Finite Probability and Applications FALL20

Academic Semester Teaching, *Boston University*
- MA113: Introduction to Statistics SPRING20

Summer Instructor, *Boston University*
- MA115,116 Statistics I-II 2015
- MA123 Calculus I 2016
- MA226 Differential Equations 2017

Teaching Fellow (graduate student), *Boston University* 2014–2020
- EK102 Linear Algebra for Engineers
- MA121,122 Calculus I for Social Sciences, Calculus II for Social Sciences
- MA123 Calculus I
- MA226 Differential Equations
- MA411 Advanced Calculus
- MA511 Analysis I
- MA569 Operations Research

Mentorship

Master student advising, (Mentee: *Lani Chastain*) AY26
- UTC: mathematics of quantum optical sensing

Undergraduate student advising, (Mentee: *Theodore Grimes, Patrick Cain*) AY25
- USMA: MA389 (Indep. Study): Numerical methods in dynamical systems

Undergraduate student advising, (Mentee: *Karly Parcell*) AY24
- USMA: MA489-499 (Senior thesis): image classification via CNN/YOLO; paper published in the ITEA journal

PhD student advising, (Mentee: *Austin Marstaller*) AY23
- Co-advised his PhD dissertation at Southern Methodist University

PhD student advising, (Mentee: *Steven Walton*) AY22
- Southern Methodist University: collaborated on a paper published in *Communications on Pure and Applied Analysis*

Undergraduate student advising, (Mentee: *William Graham*) AY20
- Directed Reading Program at Boston University: *Stochastic Differential Equations (Evans)*

Invited talks

Seminar Talk, SUNY New Paltz NOV26
"Molecular uncertainty of the nonlinear PoissonBoltzmann equation"

SIAM TX-LA, Dallas, TX OCT26
"Poisson-Boltzmann electrostatics under molecular uncertainty"

AMS Theory and Applications of Nonlinear Dynamic Equations OCT26
"Long-time continuum limits for discrete NLS"

Department colloquium, UTC SEP26
"Long-time continuum limits for twisted Schrödinger lattices"

SIAM Annual meeting, Clevaland, OH JUL26

"Complex analyticity and sparse-grid UQ for stochastic nonlinear Poisson-Boltzmann equation"

Department colloquium, Quantum Center, UTC APR26
"Magnetic flux and nonlinear quantum waves on a ring"

Department colloquium, Kennesaw State University APR26
"From lattices to waves: Bridging discrete and continuous dynamics in nonlinear dispersive systems"

AMS spring southeastern sectional, Savannah, GA MAR26
"Continuum convergence of twisted multicore fiber to magnetic NLS"

Department seminar, Kennesaw State University MAR26
"From Flach's conjecture to long-range DNLS: excitation thresholds, ground states, and stability"

Department colloquium, University of Tennessee at Chattanooga NOV25
"Excitation thresholds and breather dynamics in multi-dimensional nonlocal lattices"

AMS Fall Eastern Virtual Sectional Meeting OCT25
"Speed of propagation in nonlocal dispersive equations"

3rd UNCG Virtual PDE Conference OCT25
"Unique continuation of fractional dispersive equations"

UKC Technical Group Symposium, Atlanta, GA AUG25
"Uncertainty quantification and regularity in the stochastic nonlinear Poisson-Boltzmann equation"

SIAM Dynamical Systems, Denver, CO MAY25
"Nonlinear excitations in nonlocal lattices"

Seminar talk, University of Tennessee Chattanooga, TN APR25
"From nonlinear waves to quantum systems:
analysis, uncertainty, and applications"

Seminar talk, Creighton University, NE FEB25
"Nonlinear Waves and Quantum Systems: Analysis, Uncertainty, and Beyond"

Seminar talk, Villanova University, PA JAN25
"Bridging Nonlinear Waves and Quantum Systems: Analytical Tools, Uncertainty Quantification, and Real-World Modelss"

AMS Eastern Sectional Meeting, Albany, NY OCT24
"Kuramoto oscillators with time delay and memory effects"

SIAM Nonlinear Waves and Coherent Structure, Baltimore, MD JUN24
"Long-range Interaction on Lattice"

Invited Talk, PDE School (UC Berkeley), Berkeley, CA JUN24
"Nonlocal Discrete Solitons"

Poster, ICERM, Providence, RI APR24

"Time Evolution of the mixed-Fractional NLS"

Invited Talk, AMS East Sectional (Howard University) APR24
"Modulational instability and continuum limit of periodic fractional NLS"

Seminar Talk, CUNY Graduate Center MAR24
"Nonlocal dispersive lattice dynamics and continuum limit"

Invited Talk, AMS Special Session Mathematical Physics JMM JAN24
"On Localization of Fractional Discrete Schrödinger Equation"

Seminar Talk, SUNY New Paltz OCT23
"Application of Fractional Calculus to Physics Via Differential Equation"

SIAM NNP 2023, New Jersey Institute of Technology (Contributed Talk) OCT23
"Localization of Discrete Fractional Schrödinger Dynamics"

Great Lakes Mathematical Physics, Oberlin College (Contributed Talk) JUN23
"Discrete to Continuum Dynamics of Fractional Nonlinear Schrödinger Equation"

SIAM DS 2023, Oregon (Contributed Talk) MAY23
"Continuum Limit of 2D Fractional Nonlinear Schrödinger Equation"

12th Ohio River Analysis Meeting (Contributed Talk) MAR23
"On the Compactness of Solution Support of Fractional Dispersive Equations"

United States Military Academy West Point, New York FEB23
Teaching Demonstration

Research Talk, Navy Research Lab (Invited by Dr.Dey) JAN23
"Bifurcation of Discrete Solitons of the Nonlinear Schrödinger Equation"

SIAM 2022 TX-LA, University of Houston NOV22
"Continuum Limit of 2D Fractional Nonlinear Schrödinger Equation"

Computational Science Seminar, University of Texas at Dallas OCT22
"Continuum Limit of 2D Fractional Nonlinear Schrödinger Equation"

(Poster) SIAM Analysis of PDE 2022, Berlin, Germany (Virtual) MAR22
"Dynamics of the Mixed-Fractional Nonlinear Schrdinger Equation"

Hamilton Methods and Asymptotic Dynamics, ICERM DEC21
"Quantum Zakharov System with the Periodic Boundary Condition"

SIAM 2021 TX-LA, UTRGV (Minisymposium) NOV21
"On Mixed Nonlinear Schrödinger Equation"

Ohio River Analysis Meeting, University of Kentucky MAR20
"Periodic Quantum Zakharov System"

AMS Sectional Meeting, Tufts University MAR20
"Fourth-Order Perturbation of Cubic Nonlinear Schrödinger Equation"

Joint Mathematics Meetings (Contributed Paper Session) JAN20

"Global Well-posedness and Modified Strichartz Estimates for the Fourth-Order Schrödinger Equation"

SIAM Analysis of PDE (at La Quinta) DEC19
"Global Well-Posedness of the Adiabatic Limit of Quantum Zakharov System in 1D"

BU/Brown/UMASS Dynamics & PDE Seminar NOV19
"Pointwise convergence of Full Schrödinger Operator"

MSRI: Recent topics on well-posedness and stability of incompressible fluid and related topics JUL19
"A.E. Pointwise Convergence of Schrödinger Operator to the Identity"

Geometric and Harmonic Analysis, University of Conneticut MAR19
"Pointwise Convergence of Schroedinger Operator to the Identity"

Boston Graduate Math Colloquium, Boston College FEB18
"Nonlinear Smoothing in Nonlinear Schrödinger Equation"

BU Student Dynamics Seminar, Boston University

- "Wellposedness theory for the 1-dimensional Adiabatic Limit of Quantum Zakharov System" FEB19
- "Expository Talk: Fourier Restriction" NOV17
- "Difficulties in Extending Onsager's Conjecture to a Bounded Domain" FEB17
- "Time-Frequency Analysis and Carleson's Theorem" DEC16
- "Representation Theory Methods in Quantum Systems" NOV16

Professional Service

UTC Dept. service: Budget, Colloquium, Math Competitions, Advisory AY26–27
Article review, Chaos, Solitons and Fractals JUN26
Final judge, Interdisciplinary Contest in Modeling (ICM) APR26
Final judge, Interdisciplinary Contest in Modeling (ICM) APR25
Article review, Computers & Mathematics with Applications 203 FEB25
PhD defense committee (SMU), Austin Marstaller's PhD defense DEC24
Article review, Computational and Mathematical Biophysics 12(1) APR24
Bradley Omar Fellowship review, United States Military Academy FEB24
Grant proposal review, Army Research Office JAN24
Research Experience for Undergraduates (REU at SMU), JUL23
Article review, Communications in Contemporary Mathematics 23(4) 2020
Student Dynamical Systems Seminar at BU, Organizer 2017–2019

Skills

Machine Learning Specialization (Coursera)
(Stanford University, DeepLearning.AI, Credential Id: Q3JH7TCP34VD), AUG22
Introductory C Programming Specialization (Coursera)
(Duke University, Credential Id: GA6S3L9UMQYH) SEP22

Miscellaneous

Assistant officer-in-charge, Korean American Relations Club, USMA 2024–2025
Violin II, Boston University All-campus orchestra 2017–2018
Violin II, Boston College Orchestra 2010–2011